

# MESSIER OBJECTS

## Deep-sky catalog

The French astronomer Charles Messier (1730–1817) produced a catalog of nebulae and star clusters easily visible in small telescopes. His designations (for example M31 for Andromeda Galaxy) are still much in use today. Messier was a comet hunter (he discovered 13)

and did not want to confuse transient comets with similar-looking permanent bodies. He used a 10-cm refractor telescope in Paris, so objects south of  $-35.7^{\circ}$  declination were not included. His catalog, started in 1760, finally listed 110 objects.

| MESSIER CATALOGUE |                 |                        |                                 |                |                |                          |                            |
|-------------------|-----------------|------------------------|---------------------------------|----------------|----------------|--------------------------|----------------------------|
| Messier number    | Constellation   | Common name            | Object type                     | Messier number | Constellation  | Common name              | Object type                |
| M1                | Taurus          | Crab Nebula            | Supernova remnant               | M31            | Andromeda      | Andromeda Galaxy         | Spiral galaxy              |
| M2                | Aquarius        |                        | Globular cluster                | M32            | Andromeda      |                          | Dwarf elliptical galaxy    |
| M3                | Canes Venatici  |                        | Globular cluster                | M33            | Triangulum     | Triangulum Galaxy        | Spiral galaxy              |
| M4                | Scorpius        |                        | Globular cluster                | M34            | Perseus        |                          | Open cluster               |
| M5                | Serpens (Caput) |                        | Globular cluster                | M35            | Gemini         |                          | Open cluster               |
| M6                | Scorpius        | Butterfly Cluster      | Open cluster                    | M36            | Auriga         |                          | Open cluster               |
| M7                | Scorpius        | Ptolemy Cluster        | Open cluster                    | M37            | Auriga         |                          | Open cluster               |
| M8                | Sagittarius     | Lagoon Nebula          | Emission nebula                 | M38            | Auriga         |                          | Open cluster               |
| M9                | Ophiuchus       |                        | Globular cluster                | M39            | Cygnus         |                          | Open cluster               |
| M10               | Ophiuchus       |                        | Globular cluster                | M40            | Ursa Major     | Winnecke 4               | Double star                |
| M11               | Scutum          | Wild Duck Cluster      | Open cluster                    | M41            | Canis Major    |                          | Open cluster               |
| M12               | Ophiuchus       |                        | Globular cluster                | M42            | Orion          | Orion Nebula             | Emission/reflection nebula |
| M13               | Hercules        |                        | Globular cluster                | M43            | Orion          | De Mairan's Nebula       | Emission/reflection nebula |
| M14               | Ophiuchus       |                        | Globular cluster                | M44            | Cancer         | Beehive Cluster/Praesepe | Open cluster               |
| M15               | Pegasus         |                        | Globular cluster                | M45            | Taurus         | Pleiades/Seven Sisters   | Open cluster               |
| M16               | Serpens (Cauda) | Eagle Nebula           | Open cluster emission nebula    | M46            | Puppis         |                          | Open cluster               |
| M17               | Sagittarius     | Omega/Swan Nebula      | Emission nebula                 | M47            | Puppis         |                          | Open cluster               |
| M18               | Sagittarius     |                        | Open cluster                    | M48            | Hydra          |                          | Open cluster               |
| M19               | Ophiuchus       |                        | Globular cluster                | M49            | Virgo          |                          | Elliptical galaxy          |
| M20               | Sagittarius     | Trifid Nebula          | Emission/reflection dark nebula | M50            | Monoceros      |                          | Open cluster               |
| M21               | Sagittarius     |                        | Open cluster                    | M51            | Canes Venatici | Whirlpool Galaxy         | Spiral galaxy              |
| M22               | Sagittarius     |                        | Globular cluster                | M52            | Cassiopeia     |                          | Open cluster               |
| M23               | Sagittarius     |                        | Open cluster                    | M53            | Coma Berenices |                          | Globular cluster           |
| M24               | Sagittarius     | Sagittarius Star Cloud | Starfield in Milky Way          | M54            | Sagittarius    |                          | Globular cluster           |
| M25               | Sagittarius     |                        | Open cluster                    | M55            | Sagittarius    |                          | Globular cluster           |
| M26               | Scutum          |                        | Open cluster                    | M56            | Lyra           |                          | Globular cluster           |
| M27               | Vulpecula       | Dumbbell Nebula        | Planetary nebula                | M57            | Lyra           | Ring Nebula              | Planetary nebula           |
| M28               | Sagittarius     |                        | Globular cluster                | M58            | Virgo          |                          | Barred spiral galaxy       |
| M29               | Cygnus          |                        | Open cluster                    | M59            | Virgo          |                          | Elliptical galaxy          |
| M30               | Capricornus     |                        | Globular cluster                | M60            | Virgo          |                          | Elliptical galaxy          |